

① Business Domain

② Data collection

③ Data Understand head() tail des info()

④ Data Preprocessing

→ handle Null

→ remove duplicates

→ handle outliers

→ handle skewed

→ handle categories

→ Scaling — Normal, Stand, Robust

⑤ EDA → checking Relatn ship b/w the vari

⑥ Feature Engg → remove, add, extract → col

⑦ Split the data

⑧ Classification if data Imbalanced → Smote tech

if data balanced → Train data

⑨ Model creation

⑩ Build or train the Model

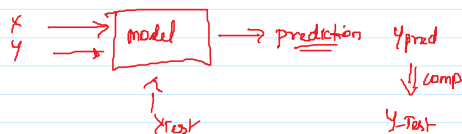
⑪ Model Evaluation

⑫ Conclude Overfitting under, General → 60-70%

⑬ hyper parameter → model, CV Evaluation, Grid Search

DT → Rad
DT → Boosting

⑭ Build the Model for hyper parameter

⑮ Overfitting & Underfitting — Bagging (All Algo)
— Boosting (DT)Loss (or) Cost function (Error) Residual = $Y_A - Y_P$

obj → To get the best fit → on what basis

Based on min error

II →

$$Y = mX + C$$

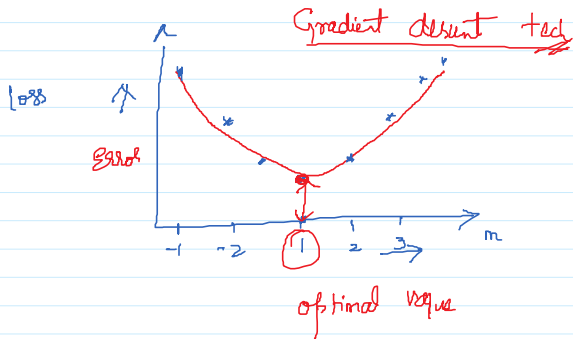
mini errors → optimal value of m & C (min error)

Gradient descent Tech

Trial		
m	C	Error
1.6	1.8	0.21
1.1	2.1	0.41

Linear Logistic Regression

m	c	Error
1.6	1.8	0.21
1.1	8.1	0.41
2	4.3	0.52
4.1	4.6	0.1



Regularization :- avoid to overfitting

Total loss (B) cost function = loss + Regularization Term

Mathematical

L1 (Lasso) L2 (Ridge)

Remove the less important features

C1 → 80%
C2 → 5%
C3 → 70%
C4 → 5%

Deep Learning :

ML



Structured

Blog the data in size

Learn the pattern

SVM, DT, KNN
LR, LogR, ...

Unstructured / structured

→ image, audio, video...

Large no. data

learn the complex relationship

NN

Neural Network :-

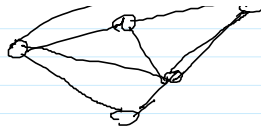
↓ neuron → brain
L1 1.



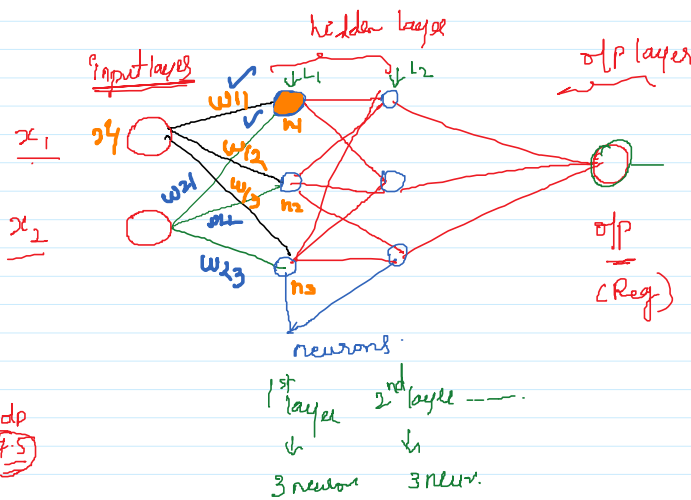
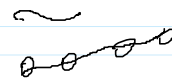
Neuron

A

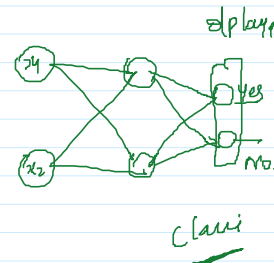
neuron $\rightarrow$ brain
 $\hookrightarrow$ function



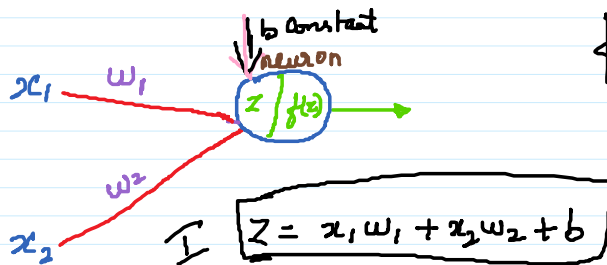
A



class



1.5 1.8 $\frac{dp}{7.5}$



$f(z) \rightarrow$ Activation function

$w \rightarrow$ weight

$b \rightarrow$ bias

$$II \quad f(z) \Rightarrow \max(0, z)$$

assume $z = \underline{1.7}$

$$f(z) \Rightarrow \max(0, 1.7)$$

$$f(z) \Rightarrow 1.7$$

o/p of the neuron $f(z) = 1.7$

Intensity

$$\max(15, 25, 7) \rightarrow 25$$



$$\text{Relu} \max(0, z)$$

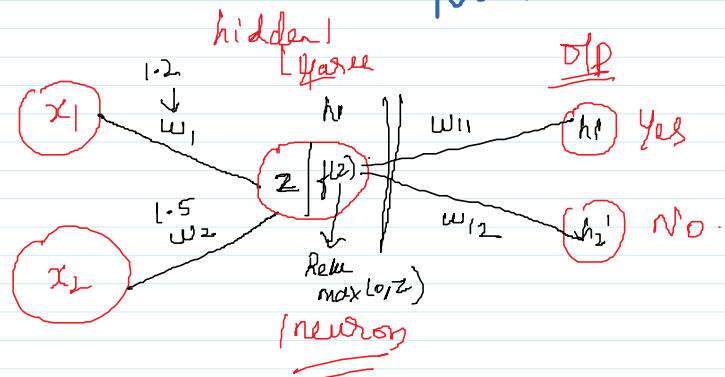
$\rightarrow +ve \text{ value}$

Feature 1	Feature 2	Target
x_1	x_2	
1.6	1.7	Yes
25.7	28.9	No

assume binary

Random value for w & b

$$\max(0, -z) \rightarrow 0$$



$$z = x_1 w_1 + x_2 w_2 + b \rightarrow 1.5$$

$$z = 1.32$$

$$f(z) = \max(0, 1.32)$$

$$zF, h, l_{new}$$